

Anatomy of an implication

When we write $p \rightarrow q$,

- p is called the hypothesis
- q is called the conclusion

We can say:

“If p , then q ”

“If p , q ”

“ q , if p ”

“ p implies q ”

etc.

All these compound statements rule out exactly one combination of values p and q

Examples

“If it is raining, then the grass is wet.”

p = It is raining (hypothesis)

q = “The grass is wet” (conclusion)

(the conclusion) says it is impossible for the situation “It is raining” and “the grass is wet” to occur simultaneously

p	q	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Related implications

“If it is raining, then the grass is wet.”

related to $p \rightarrow q$,

- we have its inverse: $\neg p \rightarrow \neg q$
 - “If it is not raining, then the grass is wet” (not how the world works)
- its converse: $q \rightarrow p$
 - “If the grass is wet, then it is raining”
- its contrapositive: $\neg q \rightarrow \neg p$
 - “If the grass is not wet, then it is not raining” (means the same as the *original*)

How to fill an implication

1. Every row where the hypothesis is false gets a T
2. Every row where the conclusion is true gets a T
3. Every unfilled row gets an F

Anatomy of a truth table

- 1st compound column list all variables in all ...
- A column for every output we are interested in

- Columns $p \rightarrow q$ & $(\neg q) \rightarrow (\neg p)$
- All their inputs occur to their left

A table compraing an implication to its contrapositive

p	q	$p \rightarrow q$	$\neg p$	$\neg q$	$(\neg q) \rightarrow (\neg p)$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

- In logic we agree that in the absence of parenthesis, we apply:
- $\neg$ first
- $\wedge$ second
- $\vee$ third
- $\rightarrow$ fourth
- $\leftrightarrow$ fifth

A bi-implication example

“You will get a grade in this course, **iff** you are registered at the end of the semester”

iff = If and only iff

Exercise

Build a table for:

$$((p \rightarrow r) \wedge (q \rightarrow r)) \leftrightarrow ((p \vee q) \rightarrow r)$$

1.3 Logical equivalences

What happens when 2 columns in a truth table are the same?

A proposition is a tautology **iff** it evaluates to T for every combination of values its variables

It is a contradiction if it evaluates to F for every combination...

It is a contingency otherwise, i.e at least one combination makes it T & at least one makes it false